

62.

Mesh Analysis

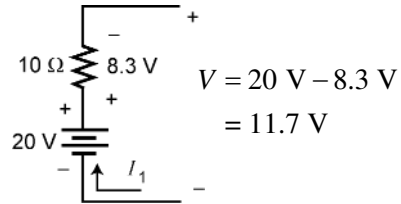
Source conversion: $E = 20 \text{ V}$, $R = 10 \Omega$

$$(10 + 10 + 20)I_1 - 10I_2 - 20I_3 = 20$$

$$(10 + 20 + 20)I_2 - 10I_1 - 20I_3 = 0$$

$$(20 + 20 + 10)I_3 - 20I_1 - 20I_2 = 0$$

$$I_1 = I_{20V} = 0.83 \text{ A}$$



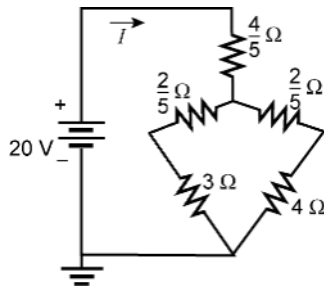
$$I_s = \frac{V}{R_s} = \frac{11.70 \text{ V}}{10 \Omega} = \mathbf{1.17 \text{ A}}$$

Nodal Analysis:

$$\begin{array}{l} \text{Node 1} \\ \text{Node 2} \\ \text{Node 3} \\ \text{Ground} \end{array} \quad \begin{array}{l} V_1 \\ V_2 \\ V_3 \\ 0 \end{array} \quad \begin{array}{l} V_1 \left[\frac{1}{10} + \frac{1}{10} + \frac{1}{20} \right] - \left[\frac{1}{20} \right] V_2 - \left[\frac{1}{10} \right] V_3 = 2 \\ V_2 \left[\frac{1}{20} + \frac{1}{20} + \frac{1}{10} \right] - \left[\frac{1}{20} \right] V_1 - \left[\frac{1}{20} \right] V_3 = 0 \\ V_3 \left[\frac{1}{10} + \frac{1}{20} + \frac{1}{20} \right] - \left[\frac{1}{10} \right] V_1 - \left[\frac{1}{20} \right] V_2 = 0 \end{array}$$

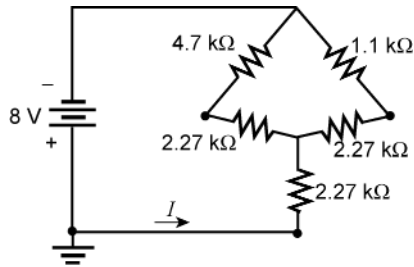
$$I_{R_s} = \frac{V_1}{R_s} = \mathbf{1.17 \text{ A}}$$

63.



$$\begin{aligned} I &= \frac{20 \text{ V}}{\frac{4}{5} \Omega + \left[\frac{2}{5} \Omega + 3 \Omega \right] \parallel \left[\frac{2}{5} \Omega + 4 \Omega \right]} \\ &= \frac{20 \text{ V}}{\frac{4}{5} \Omega + (3.14 \Omega) \parallel (4.4 \Omega)} \\ &= \mathbf{7.36 \text{ A}} \end{aligned}$$

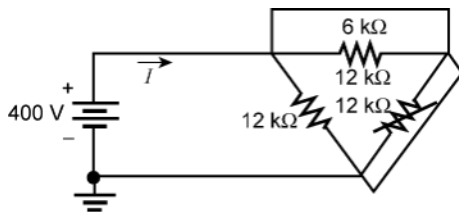
64.



$$\begin{aligned} R_T &= 2.27 \text{ k}\Omega + [4.7 \text{ k}\Omega + 2.27 \text{ k}\Omega] \parallel [1.1 \text{ k}\Omega + 2.27 \text{ k}\Omega] \\ &= 2.27 \text{ k}\Omega + [6.97 \text{ k}\Omega] \parallel [3.37 \text{ k}\Omega] \\ &= 2.27 \text{ k}\Omega + 2.27 \text{ k}\Omega \\ &= 4.54 \text{ k}\Omega \end{aligned}$$

$$I = \frac{8 \text{ V}}{4.54 \text{ k}\Omega} = \mathbf{1.76 \text{ mA}}$$

65.



(Y-Δ conversion)

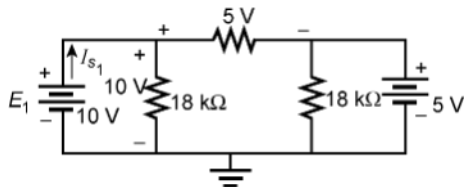
$$\begin{aligned} I &= \frac{400 \text{ V}}{12 \text{ k}\Omega \parallel 12 \text{ k}\Omega \parallel 6 \text{ k}\Omega} = \frac{400 \text{ V}}{3 \text{ k}\Omega} \\ &= \mathbf{133.33 \text{ mA}} \end{aligned}$$

66.

$$\begin{aligned} \text{a. } I &= \frac{42 \text{ V}}{(18 \Omega \parallel 18 \Omega) \parallel [(18 \Omega \parallel 18 \Omega) + (18 \Omega \parallel 18 \Omega)]} = \frac{42 \text{ V}}{9 \Omega \parallel [9 \Omega + 9 \Omega]} \\ &= \mathbf{7 \text{ A}} \text{ (Y-}\Delta \text{ conversion)} \end{aligned}$$

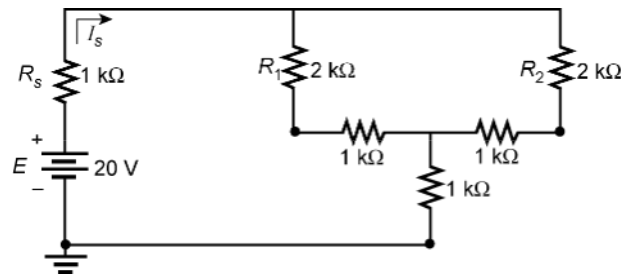
b. Δ – Y conversion

67.



$$I_{s_1} = \frac{10 \text{ V}}{18 \text{ k}\Omega} + \frac{5 \text{ V}}{18 \text{ k}\Omega} = \frac{15 \text{ V}}{18 \text{ k}\Omega} = \mathbf{0.83 \text{ mA}}$$

68. a.



b.

$$R' = R_1 + 1 \text{ k}\Omega = 3 \text{ k}\Omega$$

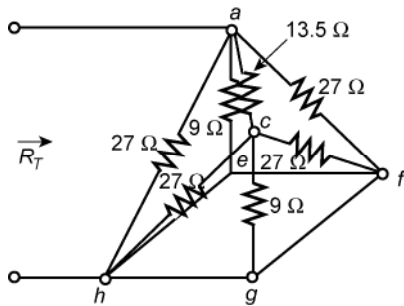
$$R'' = R_2 + 1 \text{ k}\Omega = 3 \text{ k}\Omega$$

$$R'_T = \frac{3 \text{ k}\Omega}{2} = 1.5 \text{ k}\Omega$$

$$R_T = 1 \text{ k}\Omega + 1.5 \text{ k}\Omega + 1 \text{ k}\Omega = 3.5 \text{ k}\Omega$$

$$I_s = \frac{E}{R_T} = \frac{20 \text{ V}}{3.5 \text{ k}\Omega} = \mathbf{5.71 \text{ mA}}$$

69. Using two $\Delta - Y$ conversions:



$$c - g: 27 \Omega \parallel 9 \Omega \parallel 27 \Omega = 5.4 \Omega$$

$$a - h: 27 \Omega \parallel 9 \Omega \parallel 27 \Omega = 5.4 \Omega$$

$$R_T = 5.4 \Omega \parallel (13.5 \Omega + 5.4 \Omega)$$

$$= 5.4 \Omega \parallel 18.9 \Omega$$

$$= \mathbf{4.2 \Omega}$$